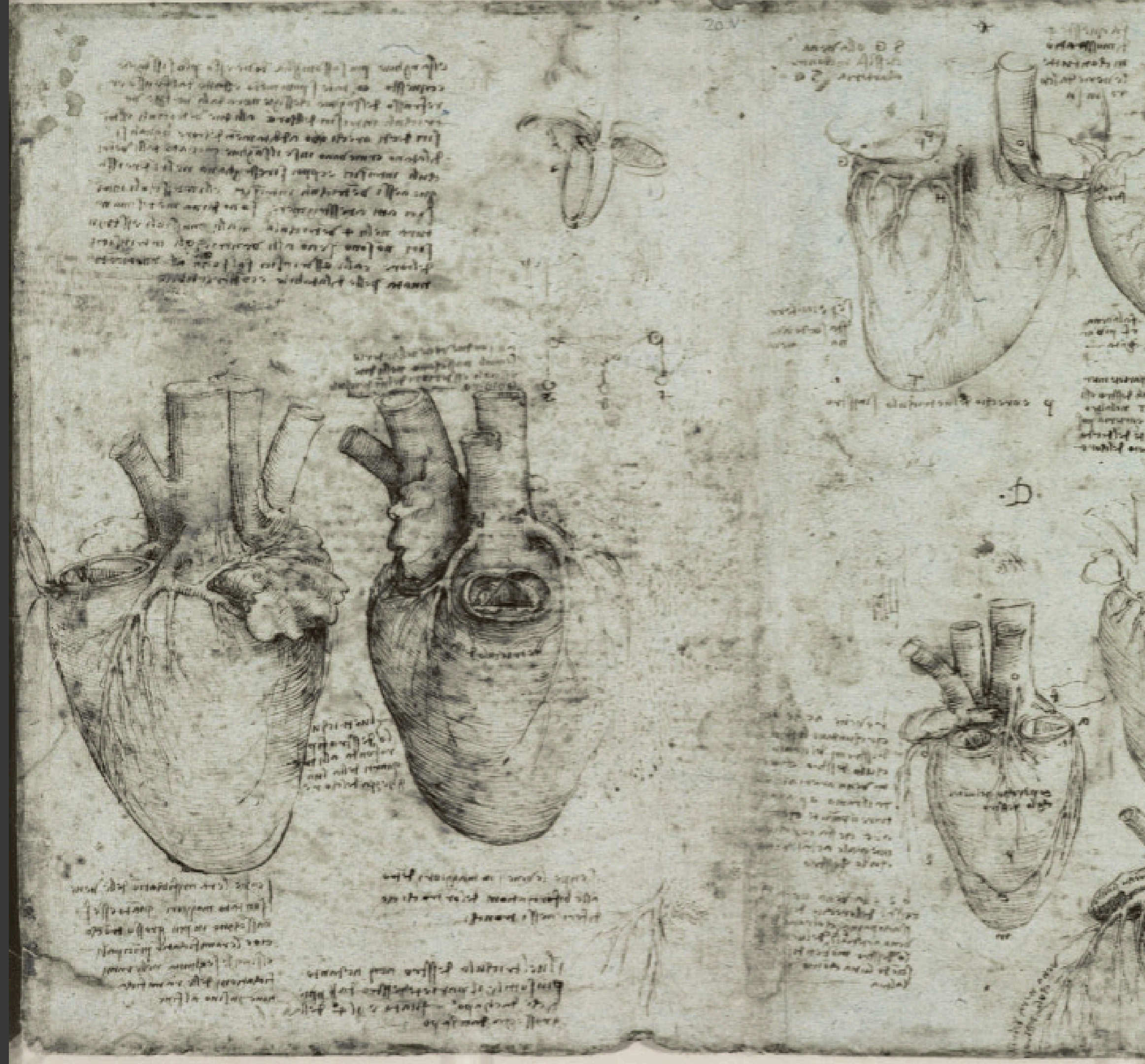


Is newer always better? Transformers against CNNs in multi-class one dimensional signal classification

Anthoulis Konstantinos mtn2501

Deep Learning: Term Project -
2nd semester 2025-2026

The heart and coronary vessels. Drawings of varying size of the heart
showing the origin of the coronary arteries and their
course; details of semilunar valves (c.1511-13). RCIN 919073 v and 919074.
Credit: Royal Collection Trust/VC Her Majesty Queen
Elizabeth II 2019.



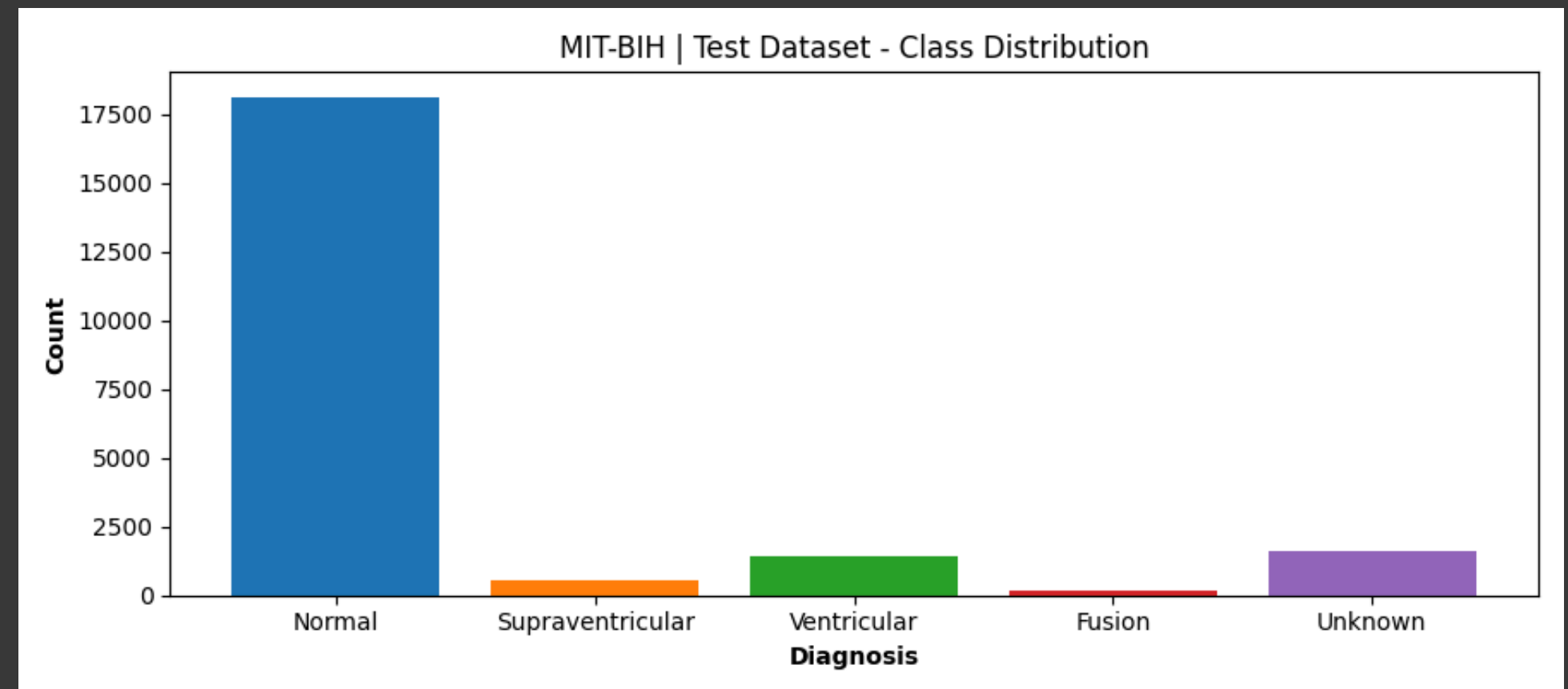
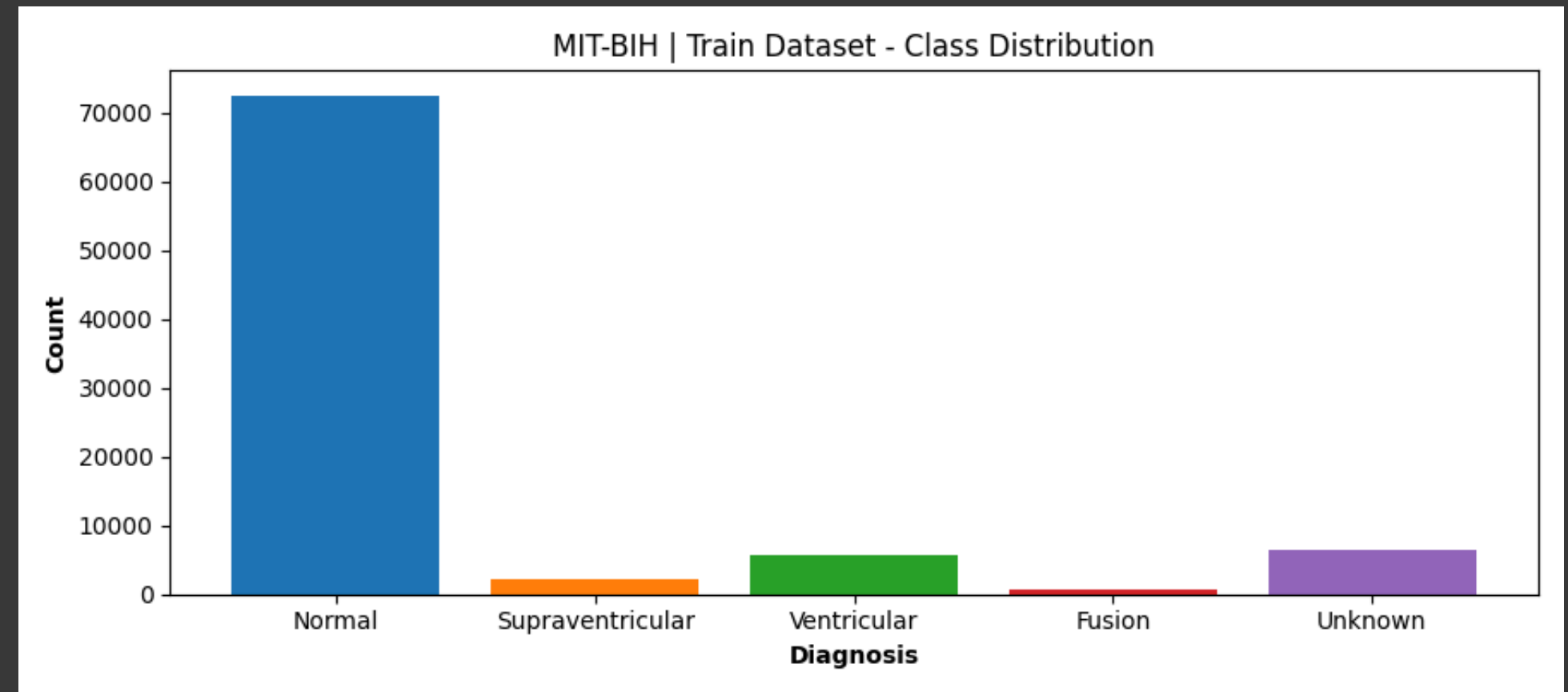
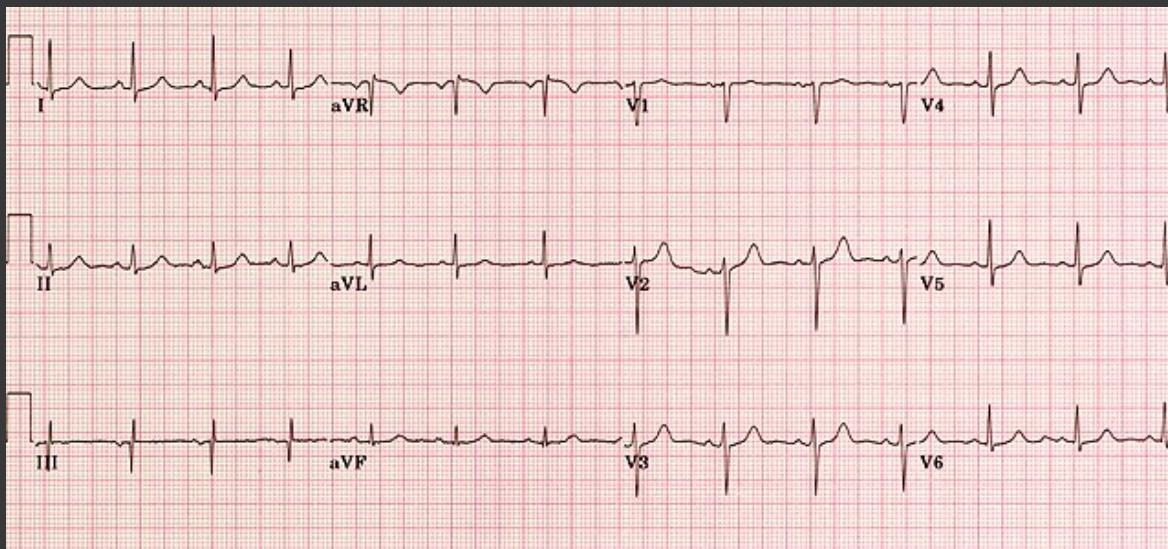
Dataset Definition

MIT-BIH ECG Dataset

A collection of ECG signals categorized into one of five different classes.

ECG: monitoring of electrical signals of the heart.

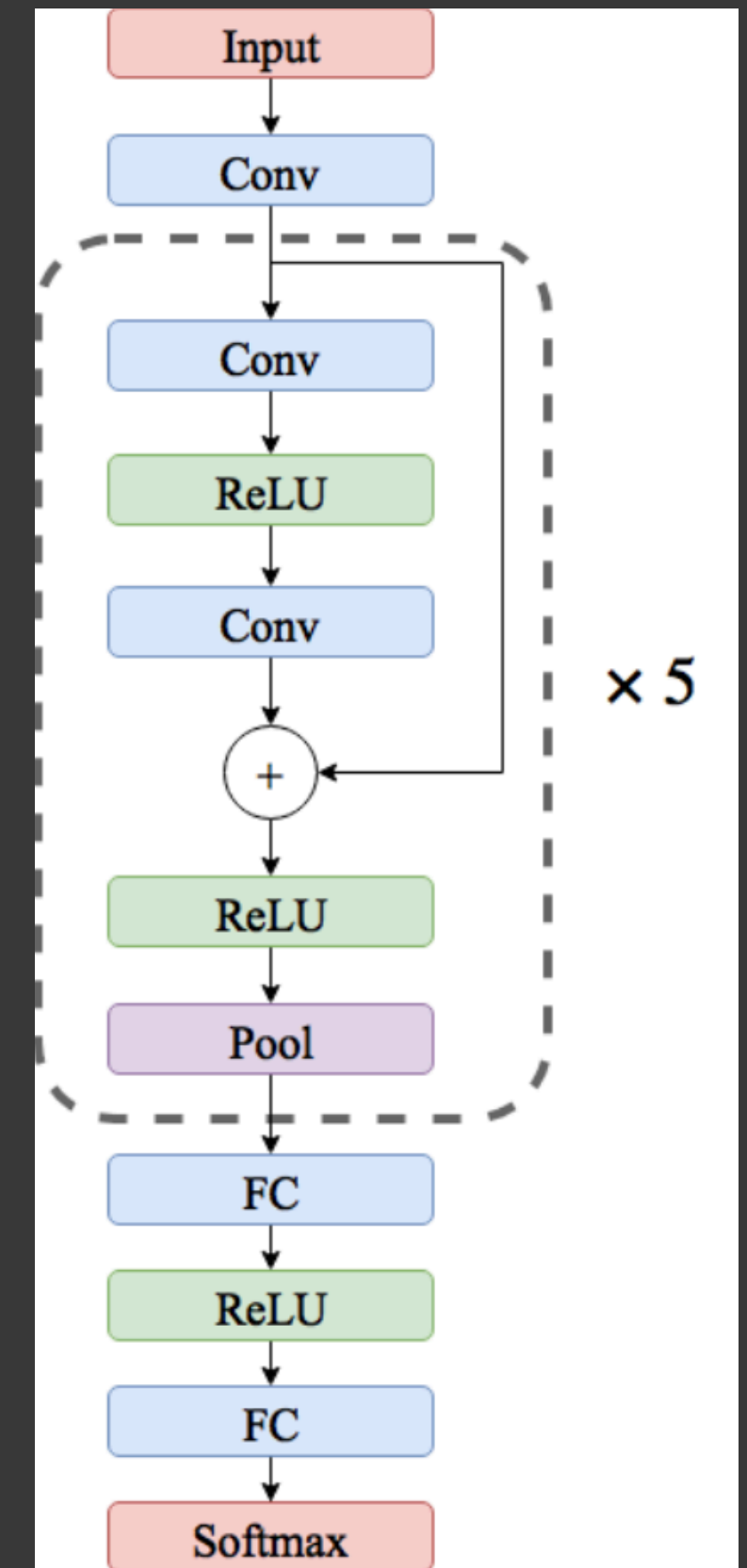
Kaggle repository has preprocessed and collated signals to the same length for ease of developing additional models.



Baseline - Convolutional Neural Networks

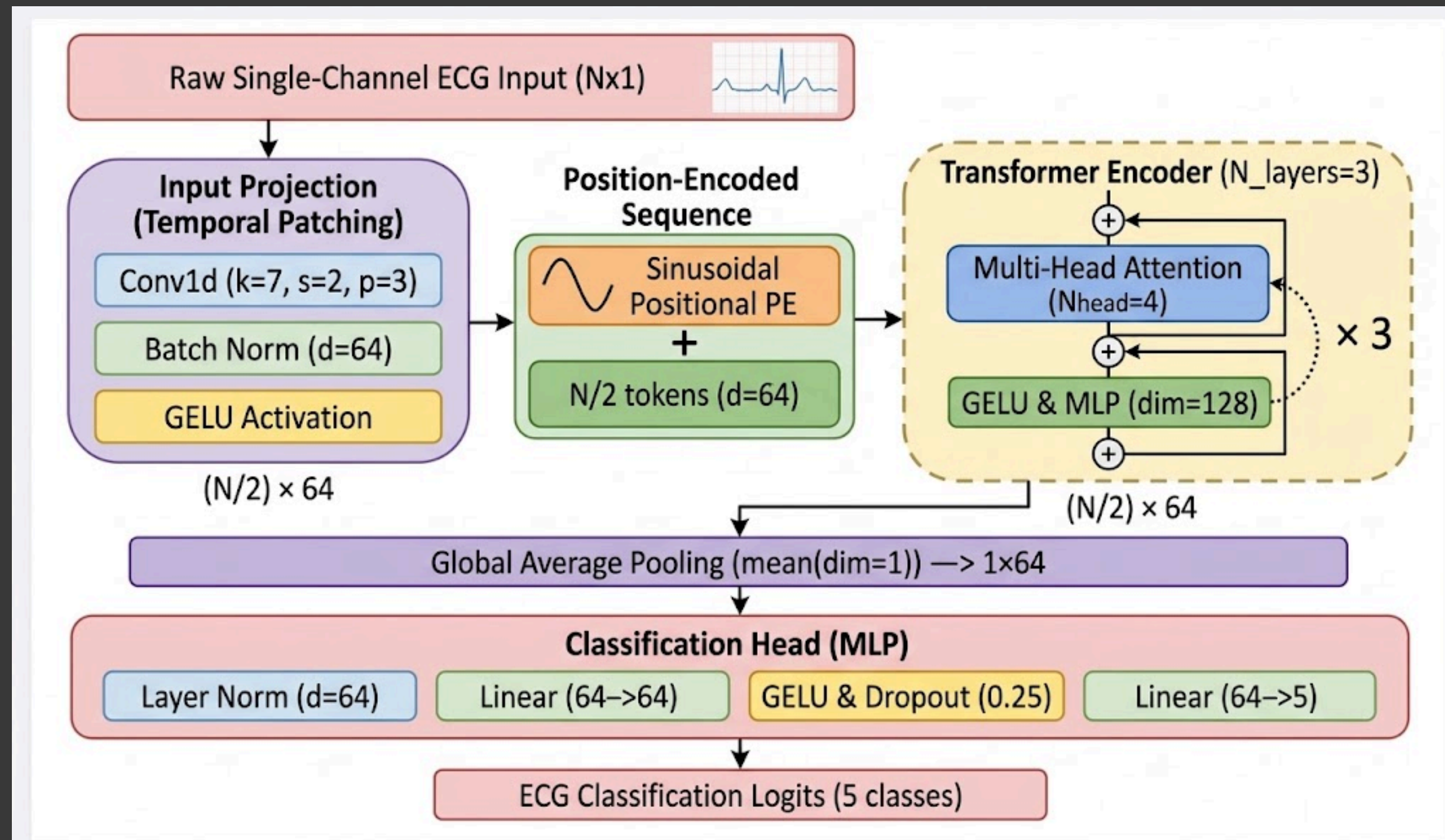
Rely on **sliding window** operations, processing the signal in subdivisions/increments before outputting the final classification result.

While computationally inexpensive and well documented, the sliding window *does not allow global processing of the entire signal.*



Transformers - Proposed Network

The proposed network accentuates convolutions with positional encodings and multi-head attention to provide both a local and global view of the signal to the transformer. The transformer output is then used as input for an MLP classifier.



Transformers - Proposed Network

Model Training Hyperparameters

Optimization & Training Loop

Model Architecture

- Embedding Dim: 128
- Attention Heads: 8
- Encoder Layers: 4
- Feed-Forward Dim: 256
- Weight Dropout: 0.1
- Out Classes: 5

- Epochs: 100
- Optimizer: AdamW
- Initial LR: 1e-3
- Weight Decay: 0.01
- LR Schedule: Cosine Annealing ($T_{\max}=100$, $\eta_{\min}=1e-5$)
- Loss: Weighted Cross-Entropy (Inverse Frequency)
- Gradient Clipping: $\max_{\text{norm}} = 1.0$
- Precision: Mixed Precision (AMP FP16)
- Early Stopping: Patience = 20 (Macro F1)

Data & Batching Setup

- Train Batch Size: 256
- Test Batch Size: 512
- GPU Vram usage: ~1GB

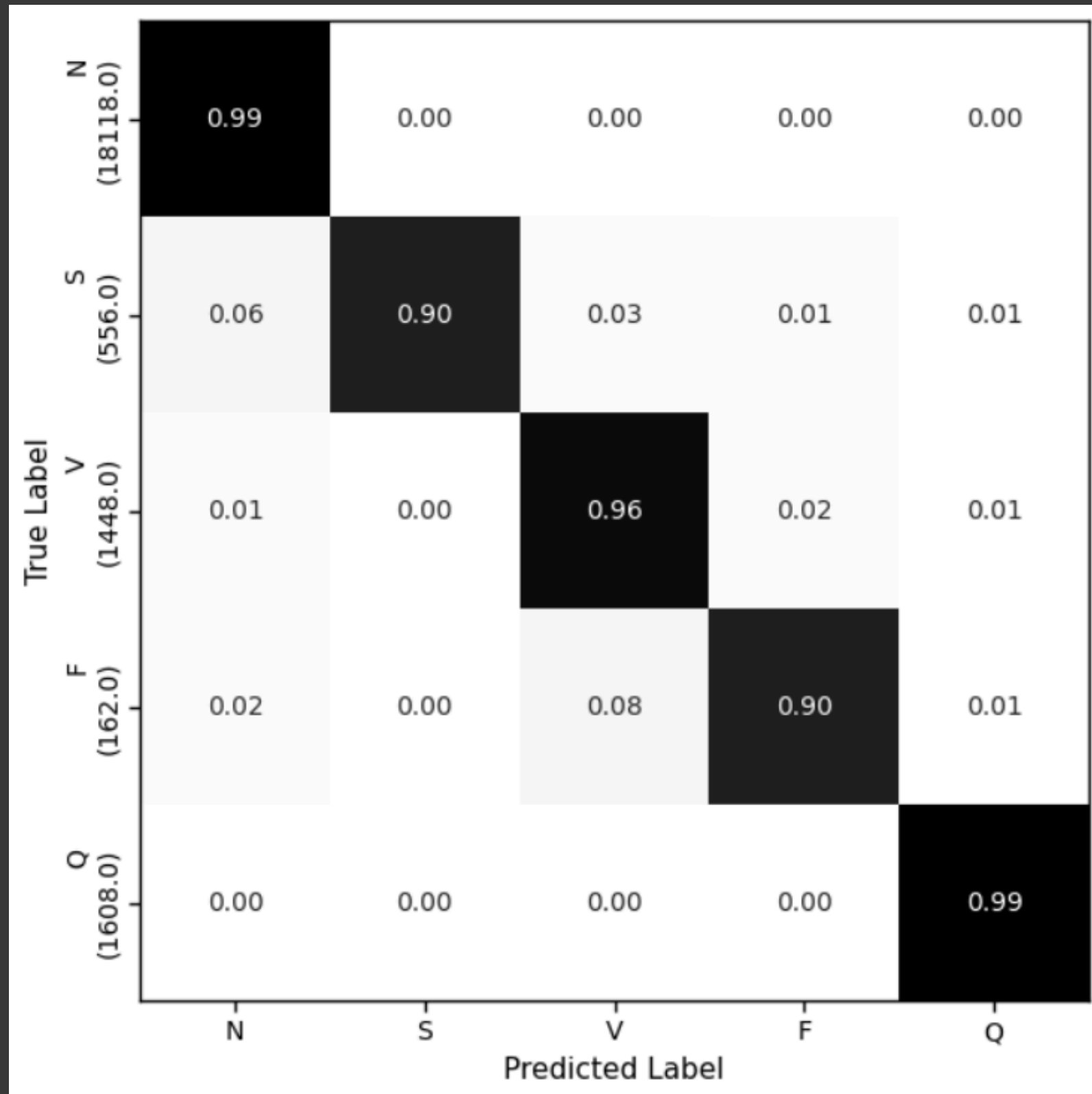
Augmentations

- Gaussian Noise Injection
- Random Amplitude Scaling
- Random Temporal Masking
- Random Cutout Masking

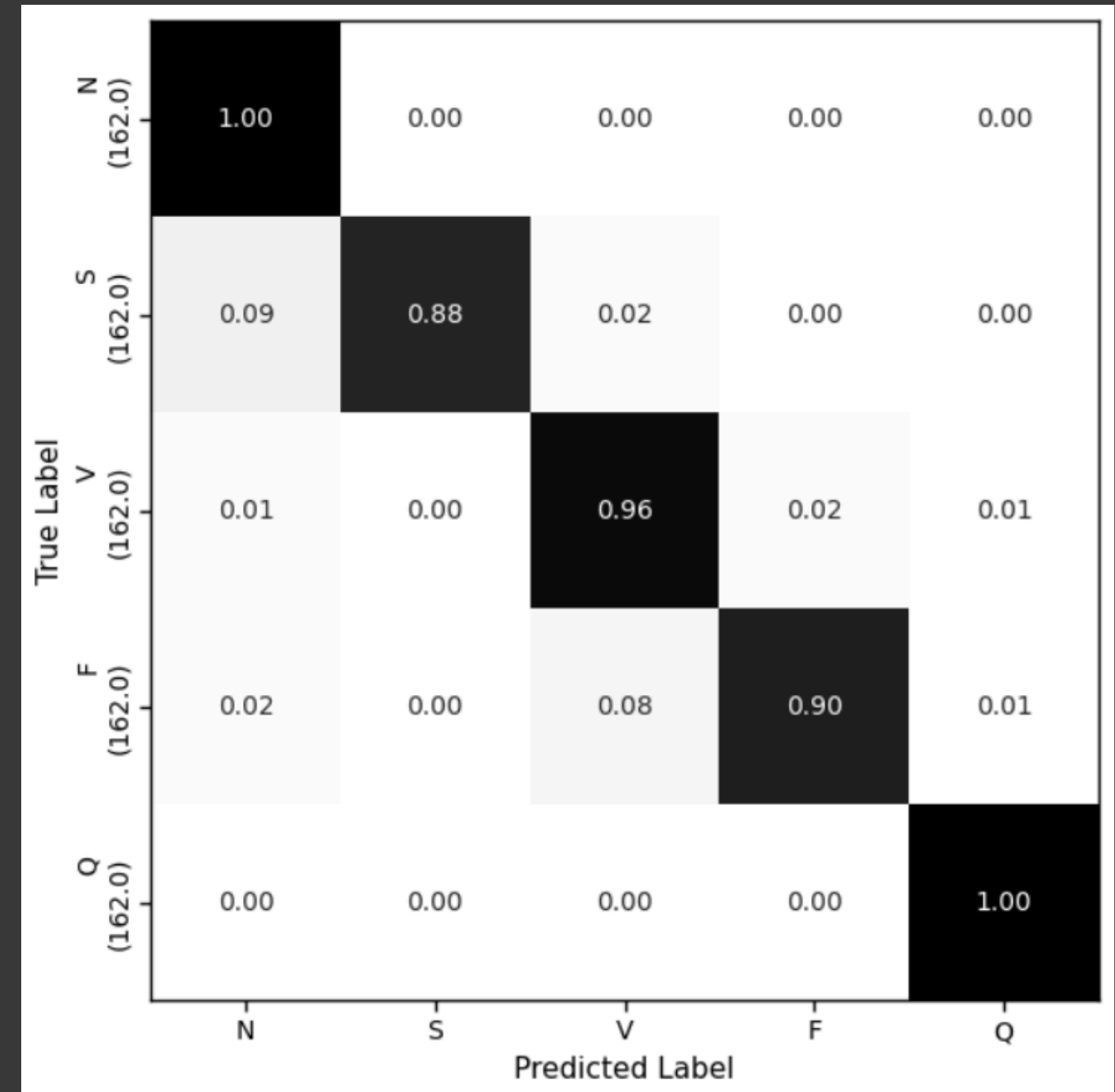
NOTE: No augmenting was applied to any samples of the *Normal* Class during training to incentivize fitting on minority cases

Training Results & Comparison with Baseline

Transformer, full test set



Transformer, balanced subset of test set

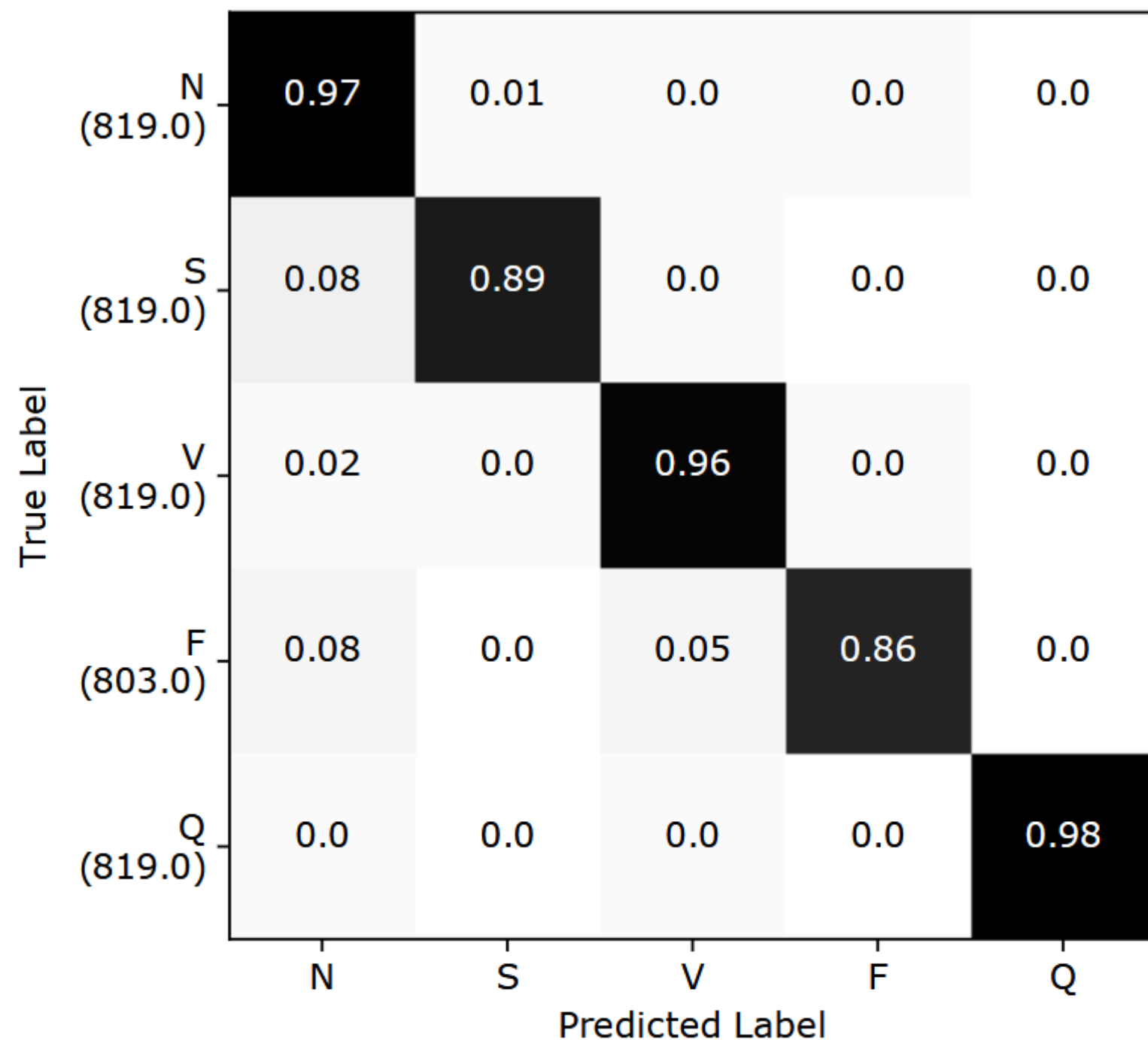


Training Results & Comparison with Baseline

CNN, balanced subset of test set. More populated than the transformer's.

NOTE

The CNN balanced test set is comprised of 819 samples from each class, while the transformer's one out of 162. This discrepancy is due to filtering the full transformer set down to the minimum number of samples present in the test set, instead of subtracting from the training data. This is purposefully done to provide the transformer with as much training data as possible. Experiments with subtracted samples from the training data led to the transformer overfit on the majority class, even with a weighted criterion in place.



Conclusions

- With proper tuning and augmentations, transformers effectively leverage large, imbalanced medical datasets to outperform CNNs.
- Minority-targeted augmentation combined with class-weighted loss successfully resolves severe data skew, achieving high sensitivity across rare pathologies.
- Additional hyperparameter tuning provides headroom for further gains, making it worthwhile to revisit benchmarks where CNNs were previously favored.